

CMPS 323 Design and Analysis of Algorithm (Quiz-2)

(Quiz is 20 minutes)

Student Name/Number: _____

Q1. Assume that you have an array of n elements. Suppose you implement Quicksort by always choosing the central element (i.e., not median) of the array as the pivot, in this case, what is the upper bound for the worst case performance?

- a. $O(n \lg n)$ ☒
- b. $O(n^2)$
- ☒ c. $O(n^4)$
- d. $O(\lg n)$

Q2. Which of the following method is the most effective for picking the pivot element for Quicksort?

- a. The element in the first position
- b. The element with the smallest value
- c. A randomly selected element
- ☒ d. The median of the elements

Q3. What is the average running time of a Quicksort algorithm?

- a. $O(N^2)$
- b. $O(N)$
- ☒ c. $O(N \lg N)$
- ☒ d. $O(\lg N)$

Q4. You must merge two sorted lists of size a and b , respectively. What is the number of comparisons needed in the worst case by the Merge algorithm?

- a. $a * b$
- b. $a + b$
- c. $\max(a, b)$
- ☒ d. $a + b - 1$

Q5. What is the worst-case time complexity of the Mergesort algorithm to sort an array of n records?

- ☒ a. $\Theta(n^2)$
- b. $\Theta(n)$
- ☒ c. $\Theta(n \lg n)$
- d. $\Theta(\lg n)$

Q6. When selecting a pivot value in Quick sort algorithm, a simple thing to do is to always pick from the same place in the partition. If we use this approach, does it matter whether we always pick from the first position in the partition, the last position in the partition, or the middle position in the partition?

- ☒ a. It is much better to pick the first value
- b. It is much better to pick the last value
- c. It doesn't really matter, they are all equally good or bad
- ☒ d. It is much better to pick the middle value

Q7. Which of the following statement is false?

- a. Divide-and-conquer algorithms should be avoided if an instance of size n is divided into two or more instances each almost of size n
- b. In the multiplication of large integers the division process is continued until a threshold value is reached
- c. In quick sort, all items smaller than the pivot are placed on the right of it ✓
- d. Divide-and-conquer algorithms should be avoided if an instance of size n is divided into almost n instances of size n/c , where c is a constant

Q8. Which of the following statement is false?

- a. We can represent a large integer by using an array of integers, in which each array slot stores one digit
- b. Strassen's matrix multiplication method requires more multiplications but fewer additions/subtractions than the standard matrix multiplication method ✗
- c. In Binary Search, the solution of the original instance is the solution to the smaller instance ✓
- d. The divide-and-conquer approach is a top-down approach

Q9. Let P be a Quick Sort program to sort numbers in ascending order using the first element as pivot. Let t_1 and t_2 be the number of comparisons made by P for the inputs $\{1,2,3,4,5\}$ and $\{4,1,5,3,2\}$ respectively. Which of the following holds true?

- a. $t_1 = 5$
- b. $t_1 < t_2$
- c. $t_1 > t_2$ ✓
- d. $t_1 = t_2$

Q10. Suppose we have a $O(n)$ algorithm that finds median of an unsorted array. Now consider a QuickSort implementation where we first find median using the above algorithm, then use median as pivot. What is the worst case time complexity of this modified QuickSort?

- a. $O(n^2 \lg n)$
- b. $O(n^2)$ ✗
- c. $O(n \lg n \lg n)$
- d. $O(n \lg n)$ ✓

Question 1. Which of the following greedy algorithms has an implementation that most resembles that of Dijkstra's algorithm?

- ☐ a. Kruskal algorithm ✗
- ☒ b. Prim algorithm ✓
- c. Huffman algorithm
- d. Floyd algorithm

Question 2. When following Huffman's algorithm, the greedy choice is to

- ☒ a. select the two probabilities of least value for merging ✓
- ☐ b. select the two probabilities of greatest value for merging. ✗
- c. add the codeword of shortest length to the code being constructed
 - d. remove the codeword of greatest length from the code being constructed

Question 3. When following Kruskal's algorithm, the greedy choice is to

- a. remove the edge of greatest cost from the graph so long as its removal does not disconnect the graph
- ☒ b. add the edge of least cost to the forest so long as its addition does not create a cycle ✓
- c. add the vertex having least connection cost to the current tree
- d. remove the vertex having greatest connection cost from the tree

Question 4. Which of the following algorithms does not require a heap for its efficient implementation?

- ✓ ☒ a. Huffman algorithm ✓
- b. Dijkstra algorithm
- ✗ ☒ c. Kruskal algorithm ✗
- d. Prim algorithm

Question 5. You are driving alone from Doha to Amman (Jordan) in a Porsche Cayenne on a fixed 2058 km route. Assume that you can go 400 km on one tank of gas and that you have a map showing every gas station on the route. What methodology for designing efficient algorithms would you choose to determine which gas stations to stop at and how much gas to get at each station?

- a. Divide and Conquer
- ☒ b. Greedy ✓
- ☐ c. Dynamic programming ✗
- d. None of the above

Question 6. Which of the following algorithm design technique is used to find the shortest path between all the pairs of vertices in a weighted graph?

- a. Greedy
- ☒ b. Dynamic programming ✓
- c. Divide and conquer
- d. Backtracking

Question 7. Which of the following statement is **FALSE**?

- a. Dynamic programming is similar to divide and conquer in that we find a recursive property
- b. In dynamic programming, the value of the optimal solution is computed in a bottom-up fashion
- ☒ c. Every optimization problem can be solved using dynamic programming ✓
- d. A divide-and-conquer algorithm for chained matrix multiplication problem is exponential-time

Question 8. Assume that there is no negative edge in a graph and you run Dijkstra's algorithm n times, once with each vertex as source. What is the time complexity in this case based on big O notation?

- ☒ a. $O(n^2)$
- b. $O(n^3)$
- c. $O(n^2 \log n)$
- d. $O(n \log n)$

Question 9. How many multiplication should we do in order to multiply an $i * j$ matrix times a $j * k$ matrix using the standard method?

- a. $i + j + k$
- b. $i * k$
- ☒ c. $i * j * k$ ✓
- d. $(i * k)^j$

Question 10. Which of the following statement is **FALSE**?

- ☒ a. We must use Kruskal algorithm when we have a graph with lots of edges ✓
- b. The time complexity of the Prim Algorithm is $\Theta(n^2)$
- c. In Kruskal's algorithm, we choose the cheapest edge that may not be in the current vertex
- d. In Kruskal's algorithm, we can put all the edges in a priority queue that is sorted by weight

Section 4: [6 pts - Solve a Recurrence Relation for a Given Problem]

Prove by induction that the solution of the following recurrence relation is $T(n) = An^2 + Bn + C$ and determine the constants, A, B, C. Assume that n is a power of 2.

$$T(n) = 4T(n/2) + n + 6, \quad n \geq 2$$

$$T(n) = 2, \quad n = 1$$